



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

Report:

Student Event Registration System

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1.0 Introduction

Student involvement in activities is important in college to enhance campus life. These offer excellent chances for education, leadership, networking, and personal development. College committees, which include student councils and clubs under the college, frequently organize events such as academic seminars, club activities, technical workshops, cultural activities, sports competitions, and charity programs. These organizations are responsible for organizing, promoting, and managing various activities to engage students and enhance the overall campus experience.

However, many of these college activities continue to use manual or semi-digital registration methods such as paper forms, spreadsheets, and basic online tools. These techniques frequently result in complications such as duplicate entries, unconfirmed registrations, quota mismanagement, and a breakdown in communication between participants and organizers. Students may lose out on activities due to an imprecise registration procedure, and the college committee may struggle to monitor an updated participant list efficiently.

To address these issues, a Student Event Registration System was created exclusively for college students and committee organizers. This system functions as a centralized portal from which students can simply discover upcoming activities, register, and receive instant confirmation. At the same time, the administrators can establish events, adjust participant limitations, view the participants' list, and track attendance more precisely and efficiently. The system organizes and stores all information digitally, which reduces errors and improves event productivity.

2.0 Problem statement

Universities often manage hundreds of student-led events annually, yet many still depend on disjointed manual or semi-digital registration methods such as paper forms, spreadsheets, or basic online surveys. These methods are not only time-consuming but also error-prone, leading to data inconsistencies, duplicated entries, and loss of participant information. Critically, these systems lack real-time validation, scalability, and integration with institutional records, making them unsuitable for high-volume or recurring event management.

This lack of centralized infrastructure creates operational friction. Students cannot access a unified event directory or verify their registration status instantly, while organizers must rely on manual quota tracking, participant filtering, and external tools for communication and updates. The absence of automation and feedback mechanisms not only reduces participation efficiency but also weakens the coordination between students and administrative units, ultimately affecting event quality and institutional engagement metrics.

Moreover, without a structured digital system, historical data on student involvement remains inaccessible, hindering meaningful event analytics and longitudinal tracking. As a result, institutions face limitations in evaluating event effectiveness, making data-driven decisions, or adapting future programs based on attendance trends. To ensure scalability, accuracy, and strategic insights, there is a critical need

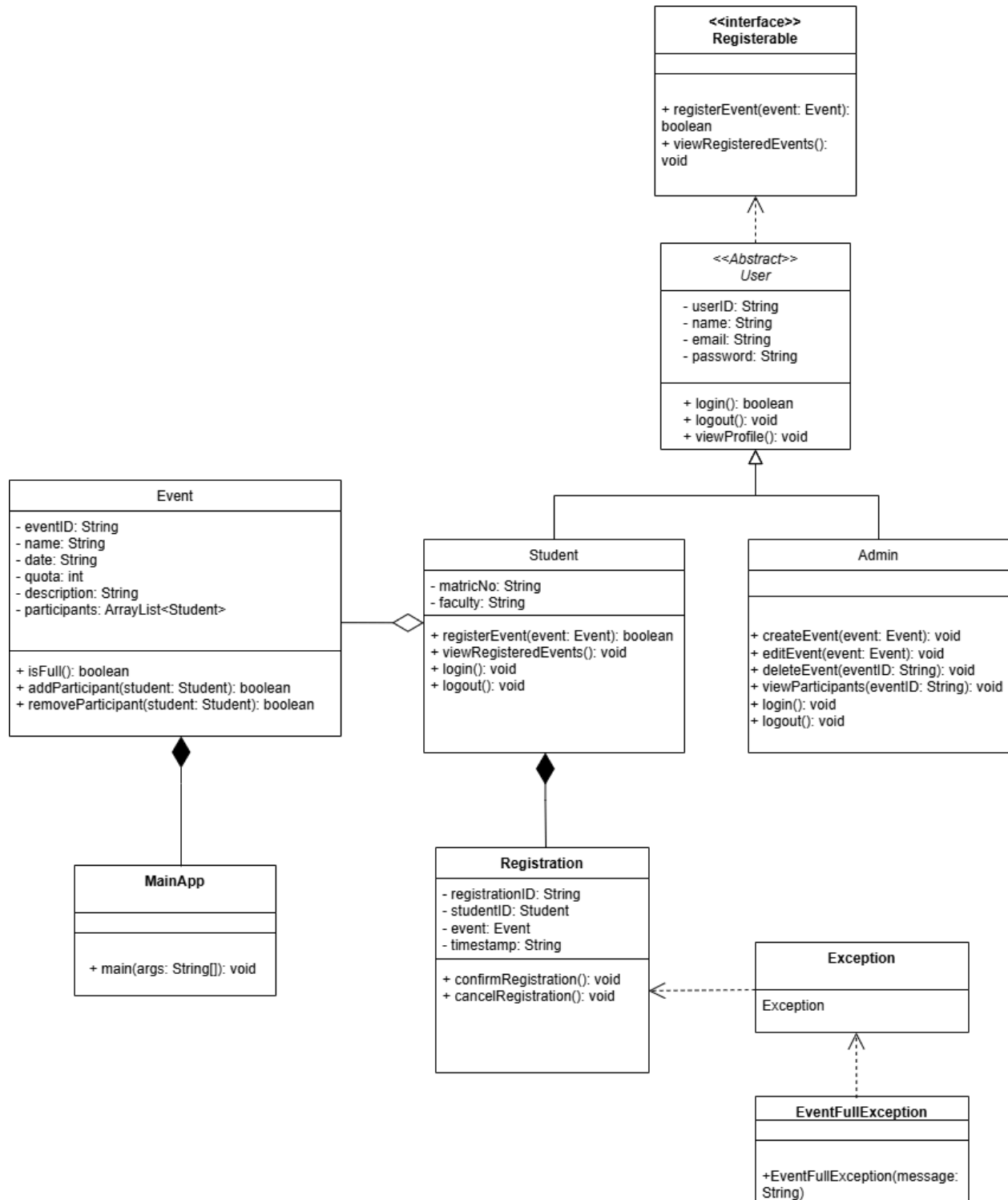
for a robust, object-oriented event registration system that centralizes management, supports real-time communication, and enhances overall user experience for both organizers and participants.

3.0 Objectives

The primary goal of this project is to create a Student Event Registration System that simplifies and digitizes the event registration process for college students. The system is designed to enhance accessibility, transparency, and efficiency in managing student involvement in various college events. Specific aims include:

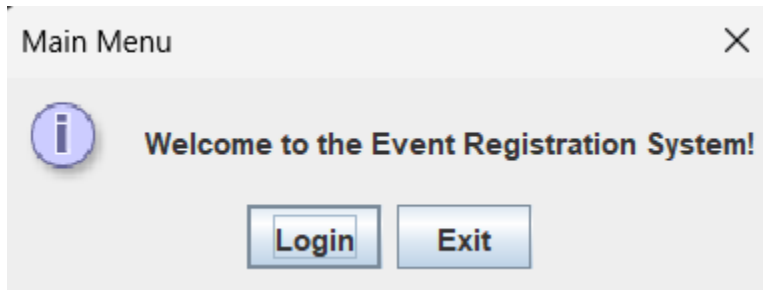
1. To make event registration easier for students by providing a centralized digital platform where they can browse available events, register, and receive confirmation immediately.
2. To assist college committees and administrators in properly planning and administering events, including setting participation limits and viewing registrant lists.
3. To minimize errors and administrative workloads associated with manual or semi-digital registration methods like paper forms and spreadsheets.
4. To create a scalable and useful Java-based graphical user interface (GUI) system by utilizing Object-Oriented Programming (OOP) concepts such as abstraction, polymorphism, inheritance, and encapsulation.
5. To increase student engagement and participation, provide real-time updates, eliminate duplicate registrations, and make the entire process more convenient and transparent.

4.0 Project Design (Class diagram)

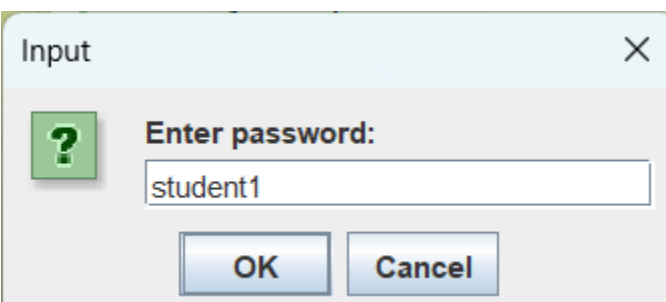
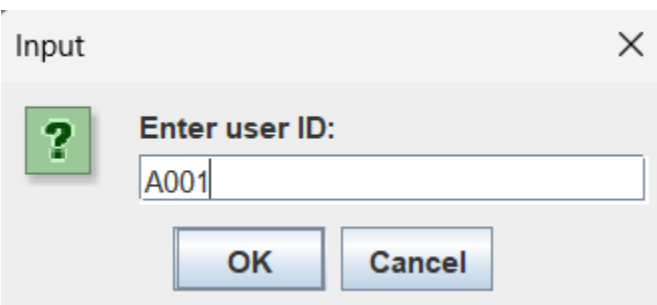


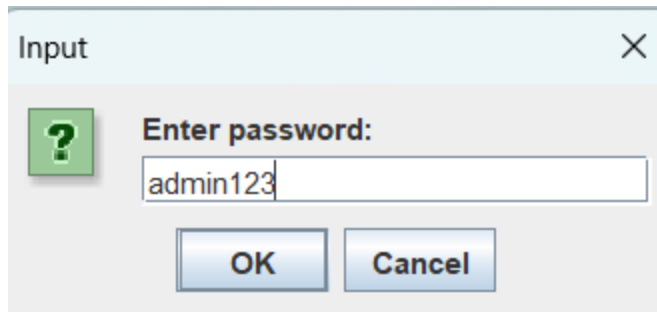
5.0 Project Output (Screenshots)

Main Menu



Login Page



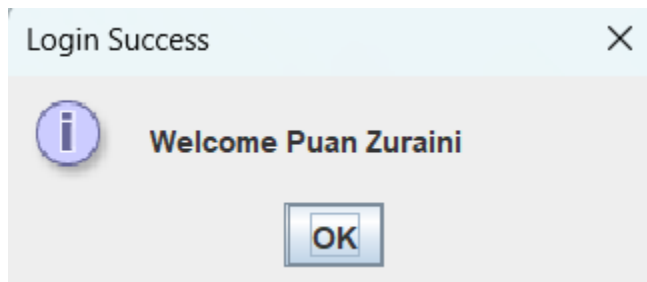


Input

 **Enter password:**

admin123

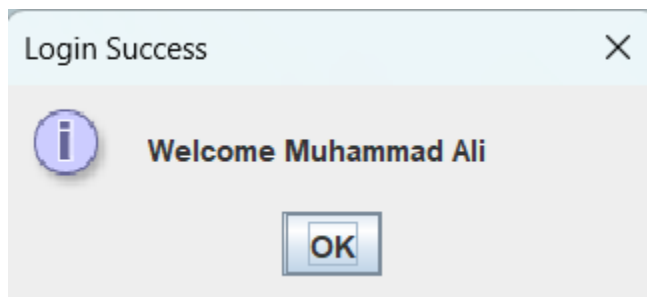
OK Cancel



Login Success

 **Welcome Puan Zuraini**

OK

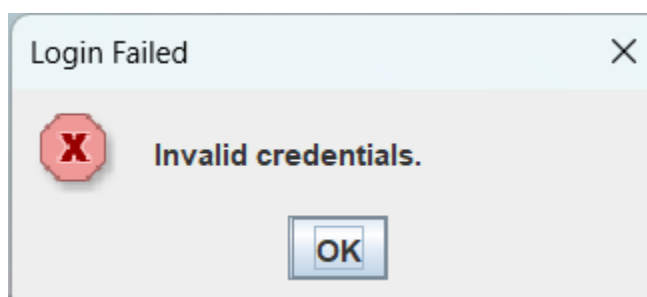


Login Success

 **Welcome Muhammad Ali**

OK

If the user enters a wrong user ID with a wrong password, it will pop up a login failed message.



Login Failed

 **Invalid credentials.**

OK

Student Menu

Student Menu

×

 Choose an option:

Register Event

View Registered Events

Cancel Event

View Profile

Logout

1. Register Event

Input

×

 Available Events:

1. Kelab Badminton KTDI (ID: EVT01) - Every Wednesday - 0/30 participants

2. Persatuan Debat (Bahasa Melayu) (ID: EVT02) - Every Friday - 0/20 participants

3. Workshop: Public Speaking 101 (ID: EVT03) - 2024-10-15 - 0/50 participants

4. Kelas Asas Fotografi (ID: EVT04) - 2024-10-20 - 0/1 participants

5. Program Sukarelawan Komuniti (ID: EVT05) - 2024-11-05 - 0/100 participants

Enter event number:

OK

Cancel

Registration Success

×

 Registered for event: Kelab Badminton KTDI

OK

If a student tries to register twice for the same event:

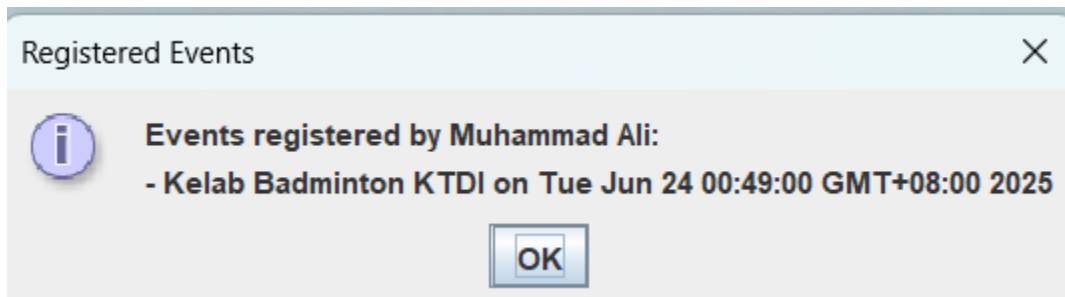
Registration Failed

×

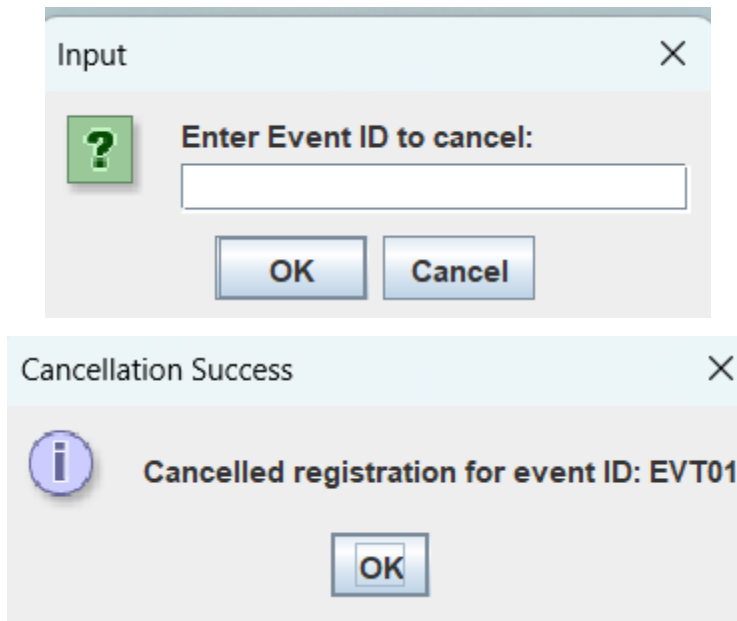
 You are already registered for this event: Kelab Badminton KTDI

OK

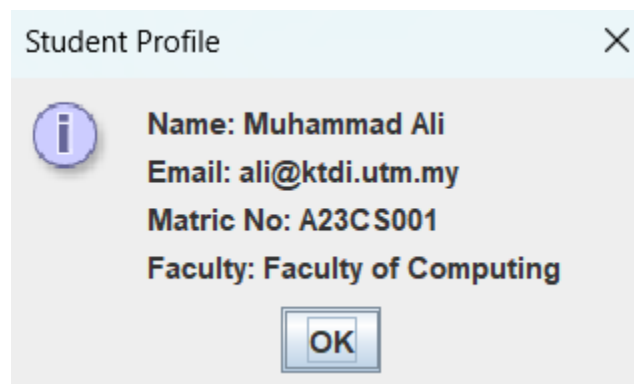
2. View Registered Events



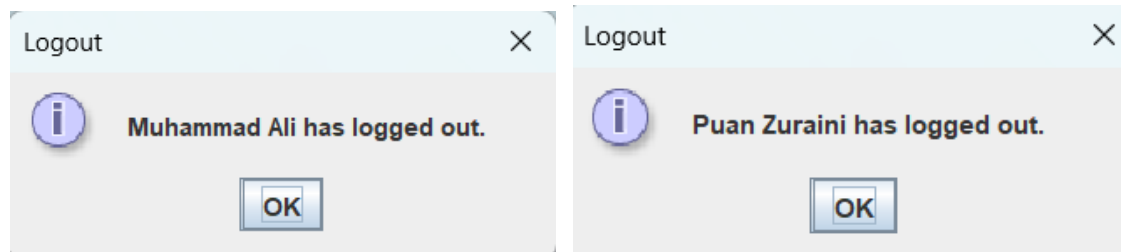
3. Cancel Event



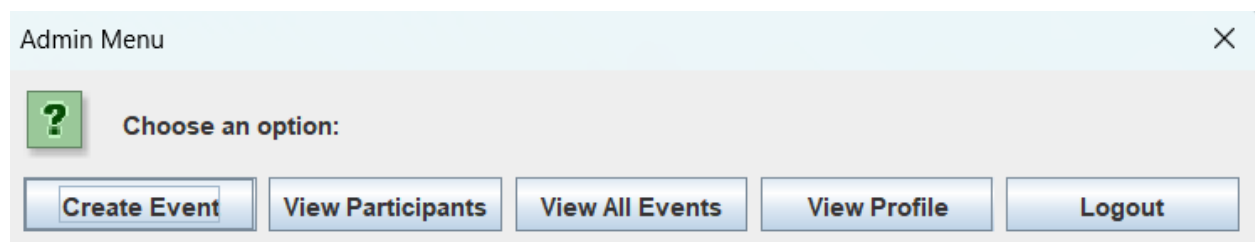
4. View Profile



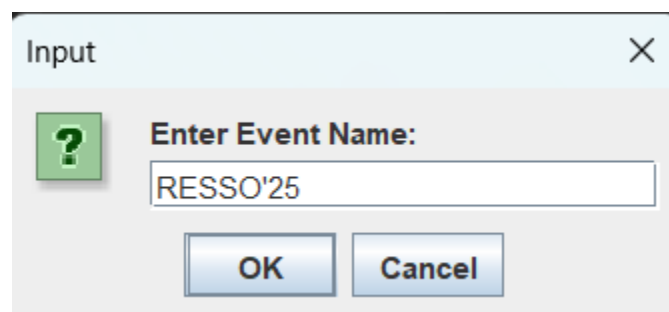
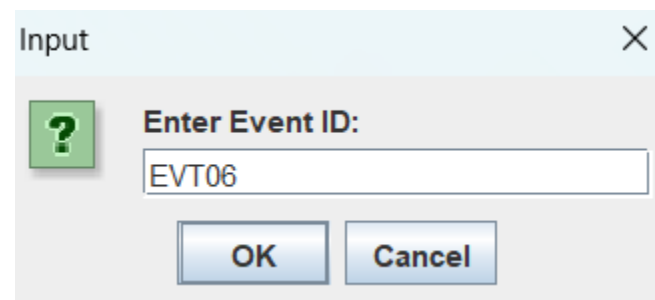
Logout page



Admin Menu



1. Create Event



Input ✕

 **Enter Event Date (YYYY-MM-DD):**

Input ✕

 **Enter Quota:**

Input ✕

 **Enter Description:**

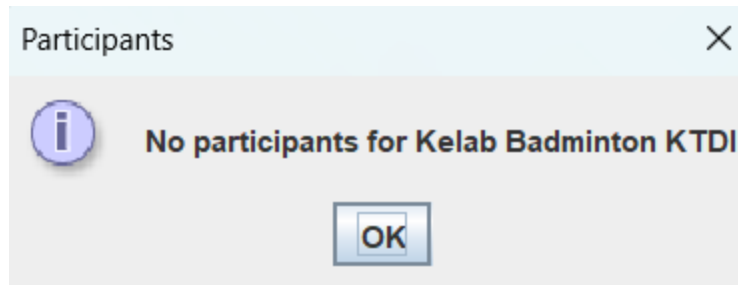
Event Creation ✕

 **Event created: RESSO'25**

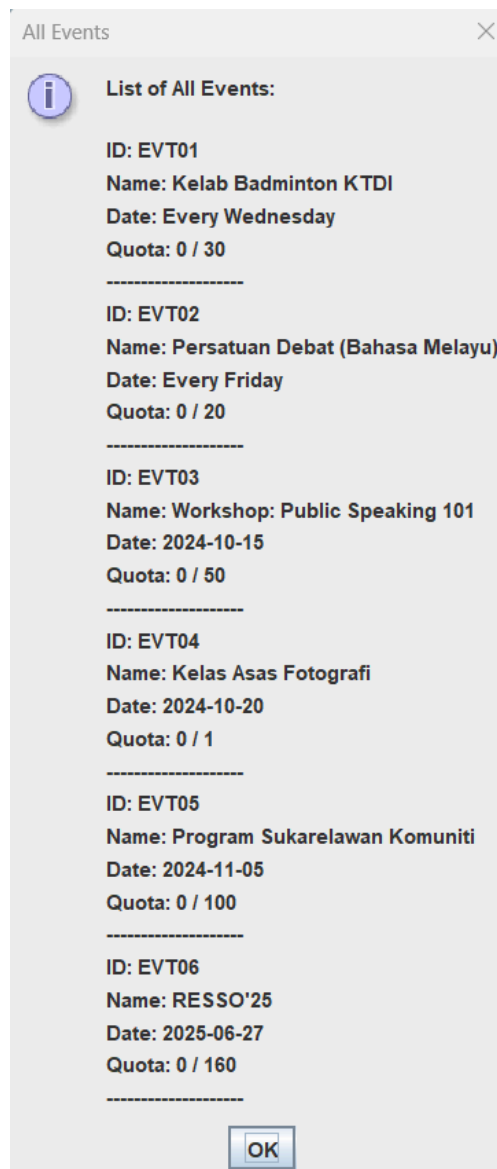
2. View Participants

Input ✕

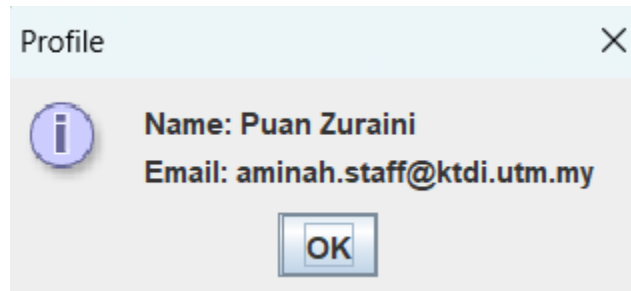
 **Enter Event ID:**



3. View All Events



4. View Profile



6.0 Conclusion

The Student Event Registration System developed in this project successfully demonstrates how Object-Oriented Programming (OOP) principles can be applied to solve real-life problems in an educational context. The system provides an efficient and user-friendly interface for both students and administrators to manage event participation in a college environment.

By automating the event registration process, the system reduces human error, eliminates redundant tasks, and ensures accurate participant tracking. Features such as login authentication, event quota management, and dynamic participant lists offer a comprehensive solution that is scalable and adaptable to a variety of student event types.

This project also reinforced core OOP concepts such as encapsulation, inheritance, abstraction, and polymorphism, along with Java Swing GUI handling via 'JOptionPane'. Future enhancements could include persistent database integration, email notifications, and mobile support, making the system even more robust and widely usable across campuses.

7.0 Appendices (Java Files)

<https://drive.google.com/drive/folders/1cZy7ZPJdf1iYRKZut8XeZAsQFHsS6RHj?usp=sharing>